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سَنَد

SANAD

امتثالٌ بدليل

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1.1 Problem definition

Organizations that process personal data in Saudi Arabia need to interpret applicable requirements and communicate their practices through privacy notices and policies.

The manual review workflow

1. **Search several legal documents**
2. **Connect a requirement to a policy statement**
3. **Decide whether the available text provides sufficient evidence**

The result: slow, and difficult to reproduce consistently.

Why a language model alone is not enough

- ! **Unsupported statements**
- ! **Misses the controlling provision**
- ! **Presents an uncertain interpretation as fact**

What a useful compliance assistant must do

1. Retrieve relevant evidence
2. Expose the source of its answers
3. Recognize when the knowledge base does not support an answer
4. **Distinguish a policy-text omission from a legally established violation**

Proposed solution

1.2

Sanad (سند)

A decision-support prototype with three connected capabilities

RETRIEVE

Retrieval-grounded answers to regulatory questions

CLASSIFY

Classification of privacy-policy segments into privacy-practice categories using OPP-115 as a labeled research corpus

COMPARE

Preliminary gap analysis comparing statements in selected Saudi privacy policies with a documented mapping of applicable PDPL requirements

1.3 Intended users and use cases

A — Privacy and compliance practitioners

Locate potentially relevant legal provisions and organize evidence for manual review.

B — Policy writers and reviewers

Identify topics that may be missing, vague, or insufficiently supported in a privacy notice.

C — Organizations and project evaluators

Inspect a repeatable workflow for regulatory retrieval and privacy-policy analysis.

D — Researchers and students

Study the application of retrieval and classification methods to privacy-policy analysis.

Primary use cases

1. **Ask a question about a PDPL requirement**
2. **Inspect cited legal passages**
3. **Classify a policy segment by privacy-practice category**
4. **Review a potential gap with the supporting policy text and regulatory reference**

2

Data card

	SOURCE	ROLE IN SYSTEM	KNOWN STATUS / NOTES
1	LEGAL **Saudi PDPL**	Primary legal source for compliance questions and requirement mapping.	Use the official SDAIA source; record the exact version and access date.
2	LEGAL **Implementing Regulation**	Detailed regulatory provisions that operationalize the law.	Official SDAIA publication; preserve article/section identifiers.
3	LEGAL **Regulation on Personal Data Transfer Outside the Kingdom**	Evidence for questions about cross-border transfer and associated safeguards.	Official SDAIA publication; preserve article/section identifiers.
4	LEGAL **Additional Saudi public policies / guidance**	Contextual policy sources; examples include data sharing, secondary data use, DPO appointment, accreditation, and HRSD Freedom of Information Policy.	Record exact title, issuing body, language, URL, version/date, and inclusion rationale.
5	EVALUATION **Team-built Gold Q&A**	Evaluation questions, reference answers, and gold legal references; includes Arabic and English items and unanswerable questions.	1. 90 items (60 Arabic · 30 English); 83 evaluated end-to-end (68 answerable · 15 unanswerable); 7 excluded.
6	RESEARCH **OPP-115**	Labeled privacy-policy corpus for privacy-practice classification.	115 website privacy policies; collected in 2015; annotated at segment level.
7	REAL-WORLD **Selected Saudi privacy policies**	Real-world material for PDPL-oriented gap analysis.	At least 10 policies planned; confirm exact sample, URLs, capture dates, and selection criteria.

7 sources · 4 legal and policy · 1 evaluation set · 1 research corpus · 1 real-world sample

3

System architecture

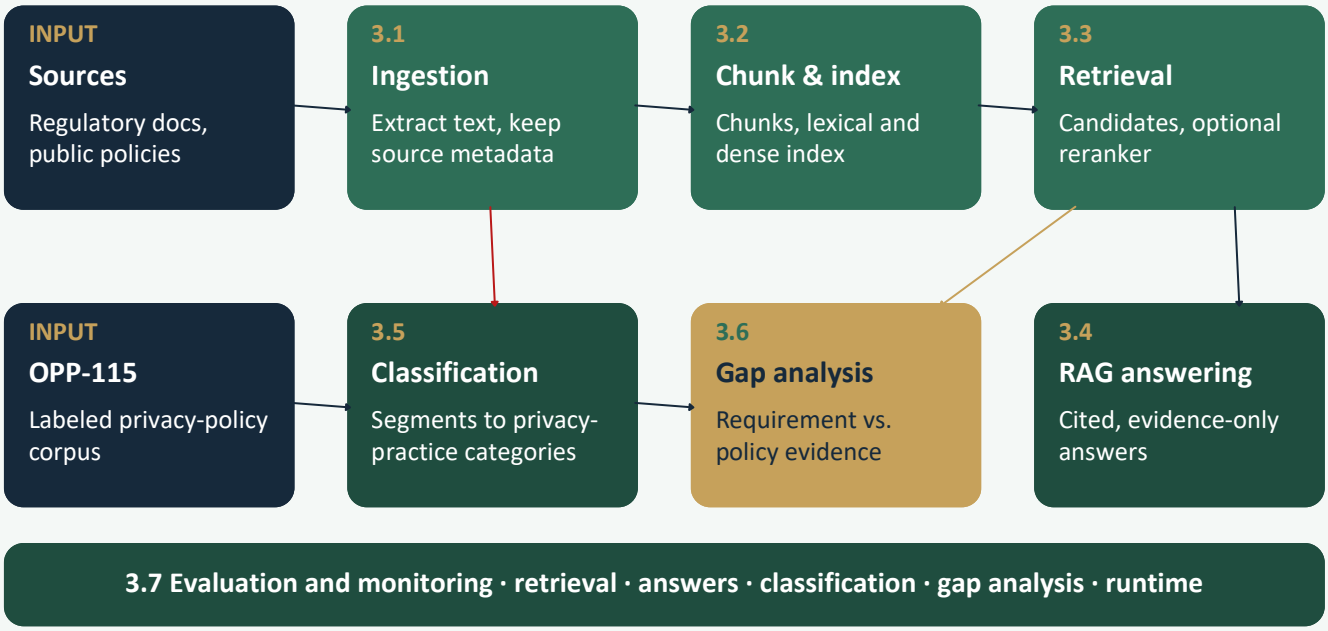


Figure 1. End-to-end architecture (as built)

3.1 Source ingestion and preparation

Collect

Official regulatory documents and selected public policies are collected with source metadata.

Extract

Text extraction preserves article, section, heading, language, URL, and document identifiers.

Normalize

Normalization supports search without overwriting the original text; keep both forms when practical.

3.2 Chunking and indexing

Documents are divided into retrievable units.: We compared both strategies and adopted article/section-aware chunks. The index combines lexical (BM25) and dense (multilingual E5) retrieval.

Article / section-aware chunks | **Fixed-size ≈ 300 tokens · 50 overlap**

Embedding limit: intfloat/multilingual-e5-base, 512 tokens max. Detect and correct oversized chunks; never truncate silently.

Chunk ID | **Document ID** | **Title** | **Source URL** | **Language** | **Section / article ID** | **Text**

3.3 Retrieval and reranking

Retrieve

Returns candidate chunks using the selected lexical, dense, or hybrid strategy.

Rerank (bge-reranker-v2-m3)

Reorders candidates by estimated relevance before context reaches the generator.

Keep metadata

Source metadata is preserved so citations can be displayed and evaluated.

3

Answering and analysis

3.4 Retrieval-augmented question answering

1. Answer only to the extent supported by the evidence
2. Cite relevant sources
3. Distinguish direct source statements from inference
4. Abstain when the available context does not answer the question

Evaluate answers and citations separately: a plausible answer is not sufficient if the supporting reference is incorrect.

3.5 OPP-115 classification

Policy segments → Features → One or more privacy-practice categories

BASELINE

A reproducible conventional model, such as TF-IDF features with a one-vs-rest Logistic Regression classifier.

MAIN APPROACH

A pretrained transformer fine-tuned for the actual label structure.

3.6 PDPL gap analysis

Requirement reference	Relevant policy excerpt, if found	Retrieval / classification evidence	Status

Evidence found | **Potential gap: no evidence found** | **Requires human review**

The output must not label an organization legally non-compliant solely because a statement was not retrieved.

3.7 Evaluation and monitoring

Retrieval quality | **Answer quality** | **Classification performance** | **Gap-analysis reliability** | **Runtime**

Each record keeps: dataset version · split · model / checkpoint · random seed · configuration · metric implementation · timestamp

AIEP GRADUATION PROJECT · TRACK B

SANAD-RAG

سَنَد

A PDPL Compliance Copilot that answers with evidence, or doesn't answer at all

«امتثالٌ بدليل» · *Compliance, with evidence*

0% Hallucinated citations	**100%** Citation validity (78/78)
93.4% Human-rated correctness	**AR | EN** Arabic and English

Not legal advice. A research assistant over the official Saudi PDPL text.

Problem, stages and knowledge base

1

Scattered obligations

The law, its Implementing Regulation, transfer rules, DPO appointment and SDAIA guidance, in Arabic and English.

2

LLMs answer confidently, and wrongly

They cite articles that don't exist and borrow from other laws such as GDPR.

3

A confident error costs more than silence

In compliance work, a wrong answer is more expensive than no answer.

The SANAD contract: every sentence is backed by a verifiable citation to an official article, or the system says so and stops.

Three stages

1

Data & Chunking

Loads 4 corpora (AR/EN × article/fixed), normalises Arabic for search only, splits Definitions into one chunk per term.

912 chunks · 0 duplicates

2

Retrieval

Benchmarks BM25, dense E5, hybrid RRF and rerank, monolingual and cross-lingual.

Recall@5 · MRR@5

3

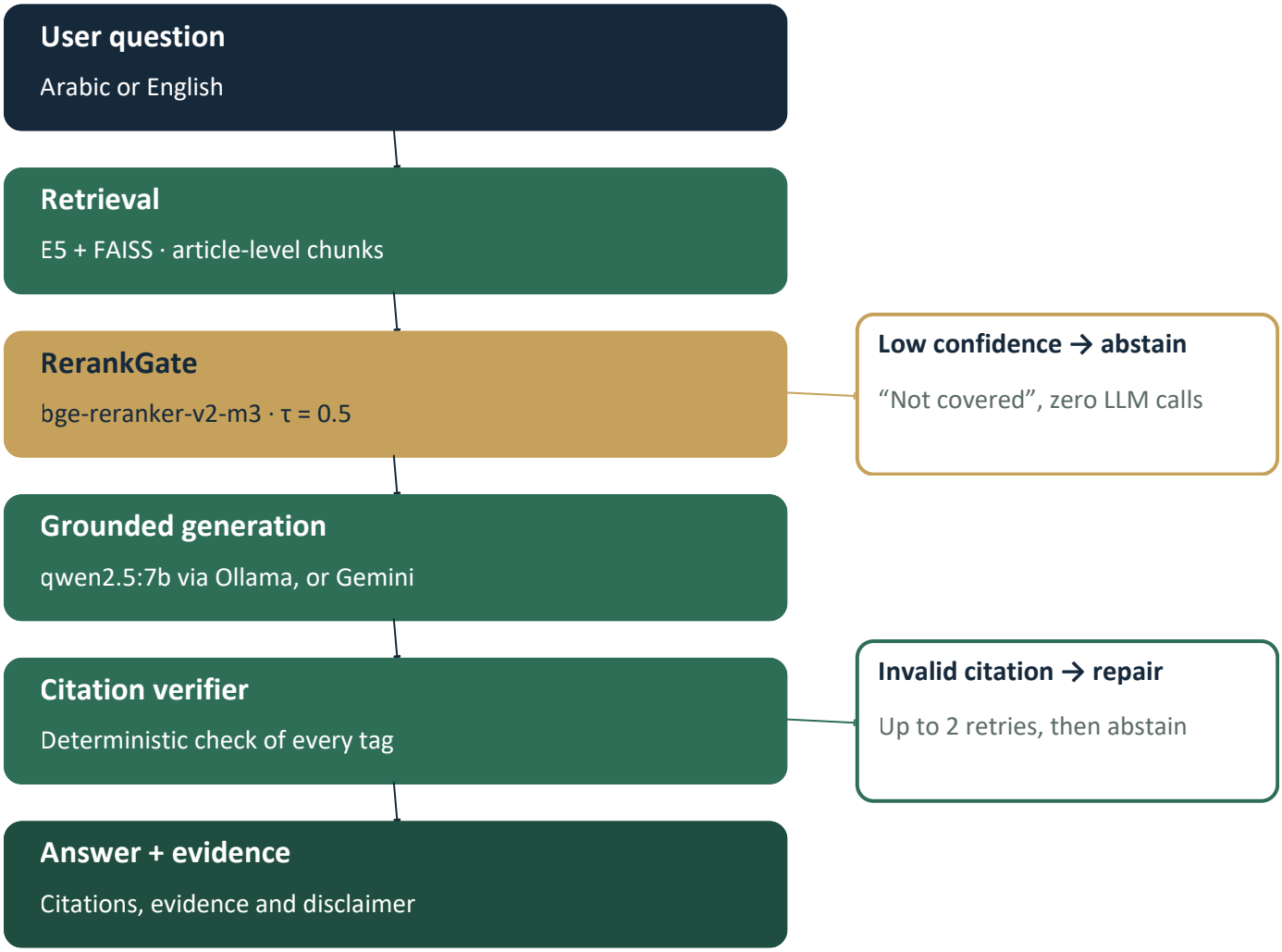
Grounded Generation

Confidence gate, generation, verification, repair or abstain. Evaluated on 83 questions.

0% hallucinated citations

912	90	83	4
validated chunks	Gold QA items	questions evaluated	corpora

Architecture and the secret sauce



We don't ask the model to be honest. We check it in code.

The secret sauce: SANAD doesn't ask. It checks.

1 Gate before you generate

A cross-encoder decides if the evidence is strong enough before any LLM call.

2 Verify after you generate

A deterministic parser checks every [Source | Article] tag against retrieved passages.

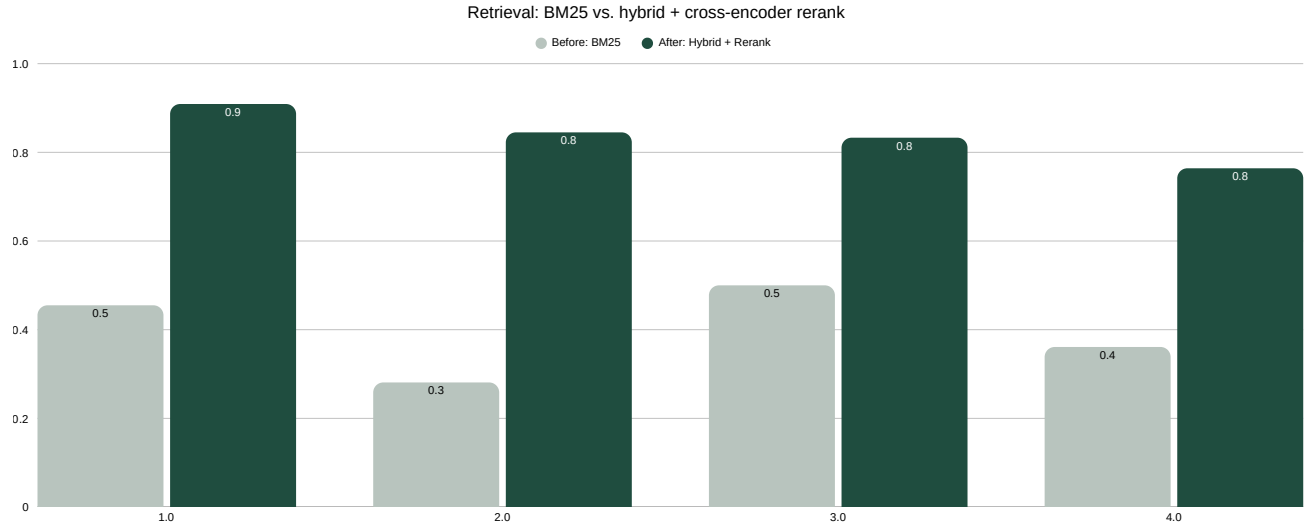
3 Repair, then refuse

Up to two repair prompts; if they fail, the system abstains.

4 Definitions first

A dedicated retrieval path returns the exact legal wording.

Retrieval and cross-lingual search

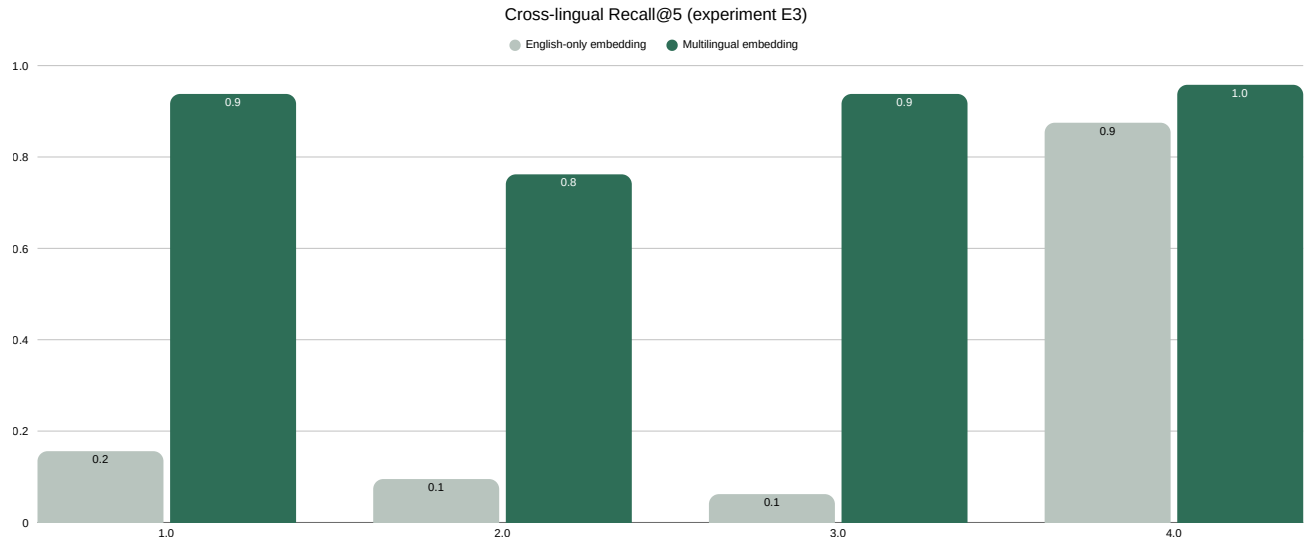


1 Arabic Recall@5 · 2 Arabic MRR@5 · 3 English Recall@5 · 4 English MRR@5

×2.0
Arabic Recall@5

×3.0
Arabic MRR@5

×1.7
English Recall@5



1 Arabic → English · 2 English → Arabic · 3 Arabic → Arabic · 4 English → English

0.06 → 0.94

An English-only embedding collapses on Arabic. The multilingual model lets an Arabic question retrieve English evidence and vice versa.

Generation and optimizations

Behaviour	LLM only	Naive RAG	SANAD
Abstains out of scope	0 / 15	0 / 15	**14 / 15 (93.3%)**
Citations to non-existent sources	25.0%	1.2%	**0%**
Valid citations	—	85%	**100% (78 / 78)**
Fabricated verbatim quotes	—	—	**0 / 63**

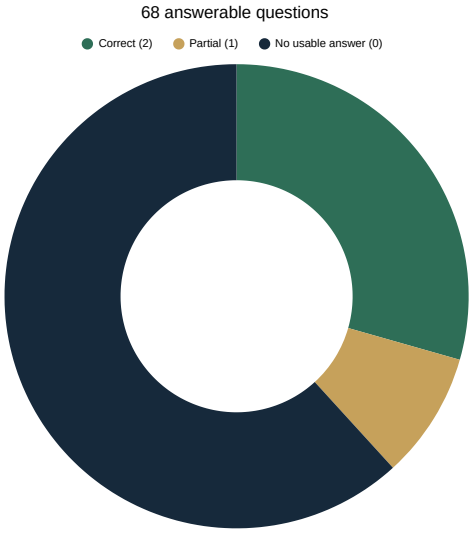
Verifier effect on one dev batch (n = 24): hallucinated-citation rate 21.4% → 0.0% after verifier + repair.

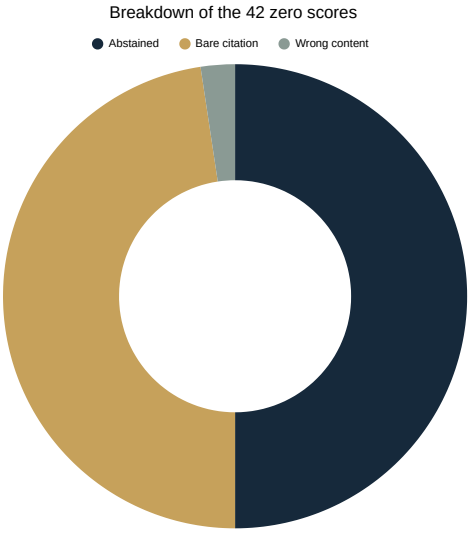
Eleven optimizations, each with evidence

1. **Article-level chunking**
   Keeps the article number for citation
2. **Definition splitting**
   One chunk per term (+19 AR · +23 EN)
3. **Arabic search normalisation**
   Original legal text never altered
4. **Multilingual embeddings**
   Cross-lingual Recall@5 of 0.938
5. **Cross-encoder reranking**
   Arabic MRR@5 from 0.675 to 0.845
6. **Language-aware RerankGate**
   0% leakage for τ between 0.2 and 0.8
7. **Suspect-question precheck**
   Stricter 0.8 gate for real-world facts
8. **Definition fusion**
   Definition chunk kept and placed first
9. **RULE 2B prompt**
   Bare citations from 20 down to 1
10. **Verifier + repair**
    Hallucinated citations 21.4% to 0%
11. **Section-level citation**
    For policy documents without articles

0 hallucinated citations
0 fabricated quotes · 0 crashes

What the human audit caught





1.5%

of answers had wrong content (1 of 68)

96%

of real answers at least partly correct (26 of 27)

14 of 21

abstentions had the right evidence in the top-5

Closing the usefulness gap

Wrongly refused answerable questions

30.9%	7.4%
BEFORE	AFTER

−76%

Bare-citation answers

20	1
BEFORE	AFTER

−95%

Answers that state an answer

39.7%	91.2%
BEFORE	AFTER

+51 pts

Definition questions AR001 · AR002 · EN001 used to abstain. They are now answered verbatim with a valid citation.

Final results and human evaluation

Metric (83 questions)	Arabic	English	All
Recall@5 (hybrid + rerank)	0.909	0.833	**0.882**
MRR@5	0.845	0.764	**0.817**
Citation validity	43 / 43	35 / 35	**78 / 78**
Fabricated verbatim quotes	0	0	**0**
Lexical evidence-support check	90.0%	76.9%	**84.8%**
Correct abstention (unanswerable)	10 / 10	4 / 5	**14 / 15**
False refusal (answerable)	3 / 44	2 / 24	**5 / 68**
Execution errors	0	0	**0**

7 / 7

Correct abstention on the held-out test split

τ = 0.5

Middle of the safe plateau, 0.2 to 0.8

93.4%

Human-rated correctness

Human evaluation of the final run: correctness

Metric	Rater 1	Rater 2	Inter-rater mean
Mean grade (out of 2)	1.81	1.93	**1.87**
Correctness (mean ÷ 2)	90.4%	96.3%	**93.4%**
Fully correct (grade 2)	60 (88.2%)	65 (95.6%)	**62.5 (91.9%)**
Grade 0 (wrong or no answer)	5	2	**3.5**

Two independent raters, on the 68 answerable questions of the final run, graded on the 0–2 rubric.

Gap analysis against 23 PDPL requirements

Requirement vs. policy evidence

The analyser reads a privacy policy paragraph by paragraph. For each requirement it decides whether the policy covers it, partially covers it, or is silent, and shows the policy sentence and the article it was measured against.

1

Split

The policy is split into paragraphs; short headings merge into the next block.

2

Classify

An OPP-115 classifier tags each paragraph with privacy-practice categories.

3

Map

Each requirement is linked to the categories that can evidence it, from an explicit checklist.

4

Judge & verify

An LLM judges the candidate paragraphs. Every quoted sentence must appear verbatim in the policy.

Verdicts

Covered

All required elements appear in mapped text, with a verbatim quote.

Partial

Some elements are present, or the wording is vague or incomplete.

Missing

No mapped paragraph or evidence addresses the requirement.

Covered means the policy mentions the element. It does not mean the organisation complies in practice.

Privacy-practice classification

Category (test n ≥ 12)	Test segments	TF-IDF + LR	Transformer
First-party collection/use	126	0.828	**0.88**
Third-party sharing/collection	93	0.794	**0.82**
International & specific audiences	30	**0.949**	0.93
User choice/control	30	**0.800**	0.78
Data security	19	0.727	**0.80**
User access, edit & deletion	14	**0.839**	0.78
Policy change	12	0.846	**0.87**
Other	96	0.683	**0.72**

3,729

Segments kept when two of three annotators agree (63 dropped)

92 / 11 / 12

Policies in train / validation / test, split by policy

0.809

Baseline macro-F1 on test (95% CI 0.76 to 0.87)

Rare categories and aggregate scores

Measure	TF-IDF + LR	Transformer
Macro-F1	0.809	**0.839**
Macro-F1 (0.5 Threshold)	**0.831**	0.830
Micro-F1	0.79	**0.82**
Confidence Interval	0.76 – 0.87	**0.796 – 0.907**

Per-label accuracy is high because a segment carries about 1.15 of 10 labels, so macro-F1 ranks the models. At the default 0.5 threshold the baseline reaches macro-F1 0.831; the rare categories are not interpreted individually.

Reading the results, and their limits

1

Vocabulary decides the ranking

Categories with distinctive wording (international audiences 0.949, policy change 0.846, access and deletion 0.839) score highest. “Other” (0.683) is a heterogeneous catch-all, and many of its errors are short headings such as “Information You Provide to Us”.

2

Transformer effect

TO CONFIRM: report, per category, whether the fine-tuned transformer improved on the baseline, and whether the 95% interval of the macro-F1 difference excludes zero. If it includes zero, the difference is not established on this test set.

3

Why macro-F1, not accuracy

Per-label accuracy (0.953) is inflated because a segment carries only 1.15 of 10 labels on average. Macro-F1 (0.809) is the headline figure. The predicted label set matches the gold set exactly for 64% of segments, and at least one label is right for 87%.

4

Error analysis

A short heading such as “Information You Provide to Us” is labelled Other. A “Choice/Opt-Out” segment labelled Other was predicted as user_choice_control, which may be annotation noise rather than a model error.

5

Small test set, unstable scores

The test set holds 12 policies: Data retention has 6 segments and Do not track 3, so their scores are not reliable. Fine-tuning also varies across random seeds, so small gaps count as inconclusive.

What was hard, and how we solved it

1

Getting the data to load

Three OPP-115 files had paths longer than 260 characters and would not open on Windows, so we built a clean copy without original_policies. The policy ID inside the annotations also differed from the file-name prefix; we matched file names against policies_opp115.csv.

2

Annotators disagree

Three annotators labelled each segment, and we kept a category only when at least two agreed. This cut multi-category segments from 52% (union of annotators) to 18%, and dropped 63 segments (1.7%).

3

Imbalanced categories

Do not track has only 3 test segments and Data retention has 6, so their scores are not reliable. We report them separately and do not interpret them individually.

4

Thresholds on a small validation set

Per-class thresholds tuned on 11 validation policies gave macro-F1 0.809, below the default 0.5 threshold (0.831). Data retention fell from 0.91 to 0.62 after tuning on only 9 examples.

5

Compute and domain shift

Fine-tuning the transformer needed a GPU (Colab), and long segments are truncated at 256 tokens. Training data is US English web policies, so results on Saudi policies need caution.

Future work

Directions beyond the current prototype

Answer quality

Stronger retrieval for short questions and definitions, and a model that cites consistently.

Wider knowledge base

Add SDAIA guidance and other regulatory sources, and refresh them when the law changes.

Arabic classification

Train a classifier that reads Arabic policy text directly, instead of relying on an English model.

Citation verification

Check that the cited article supports the claim, not only that it exists in the retrieved text.

Policy comparison

Show differences between organisations, and between a policy and the requirements for its sector.

Human review loop

Let specialists correct errors and feed those corrections back into retrieval and judging.

Public deployment

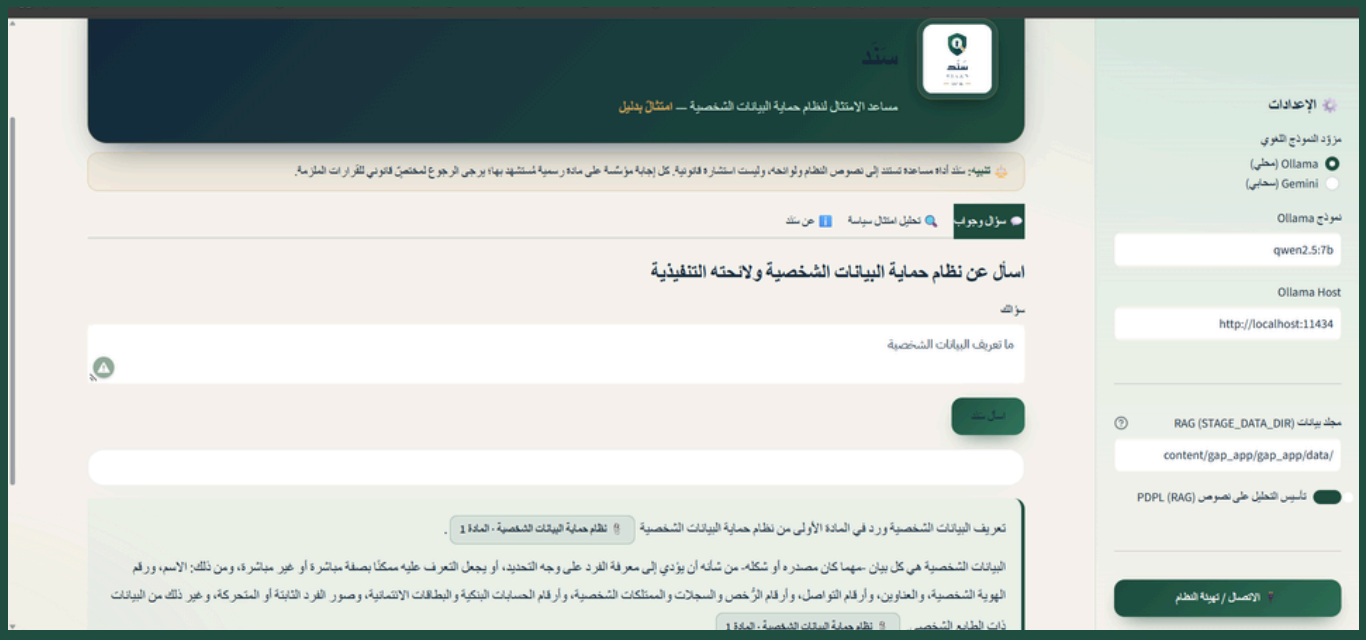
A stable public link with key management, usage limits, and performance monitoring.

Legal change tracking

Flag when the law or its regulations change, and list the requirements affected.

Appendix

Figure C.1: Answer to a definitional question (“What is personal data?”), showing the citation badge for Article 1.



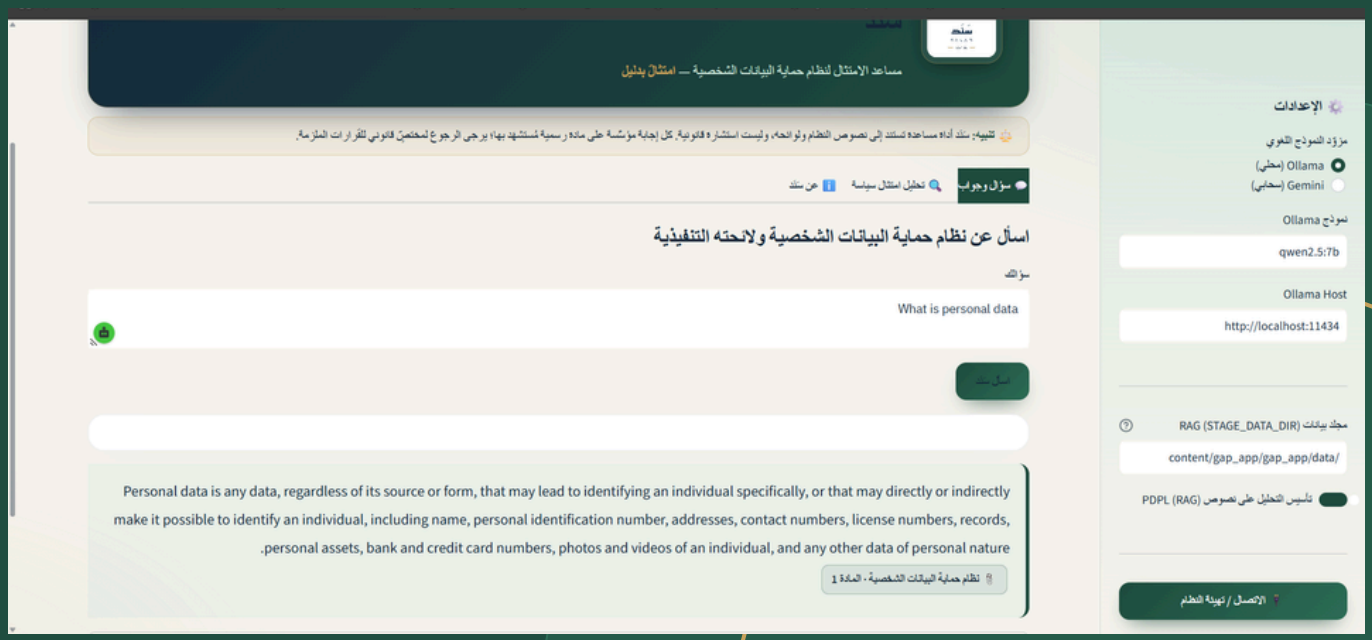
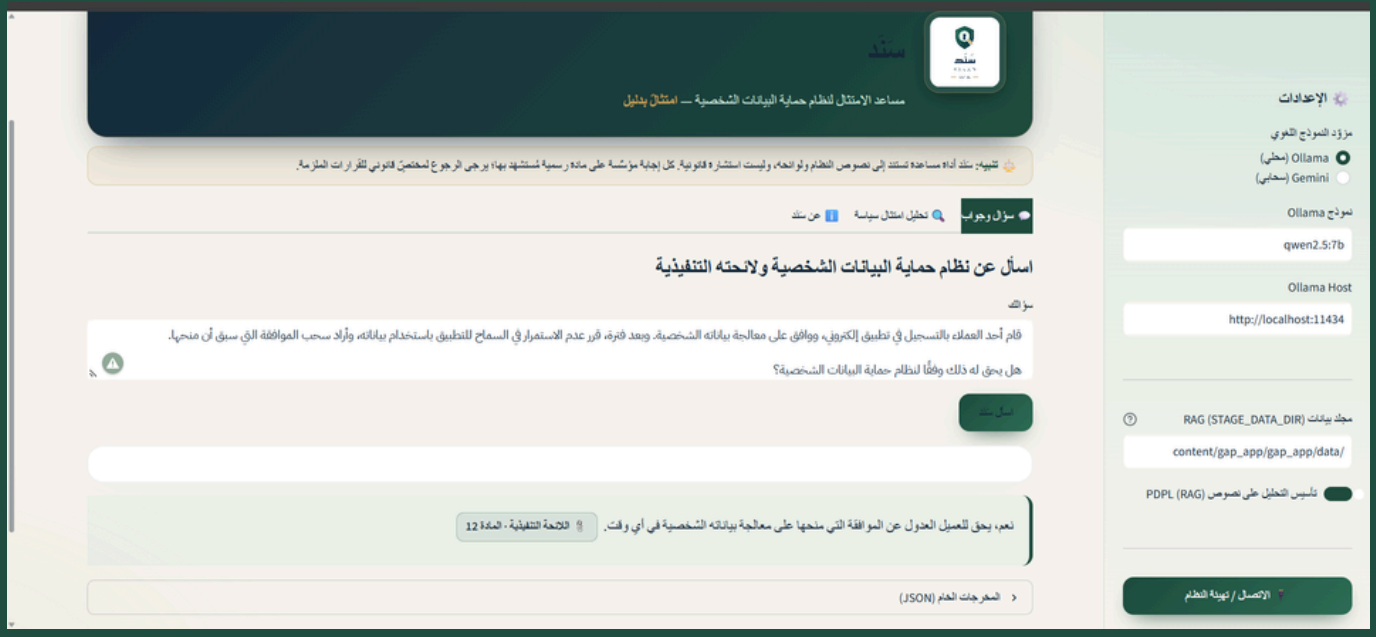


Figure C.2: Answer to a practical scenario



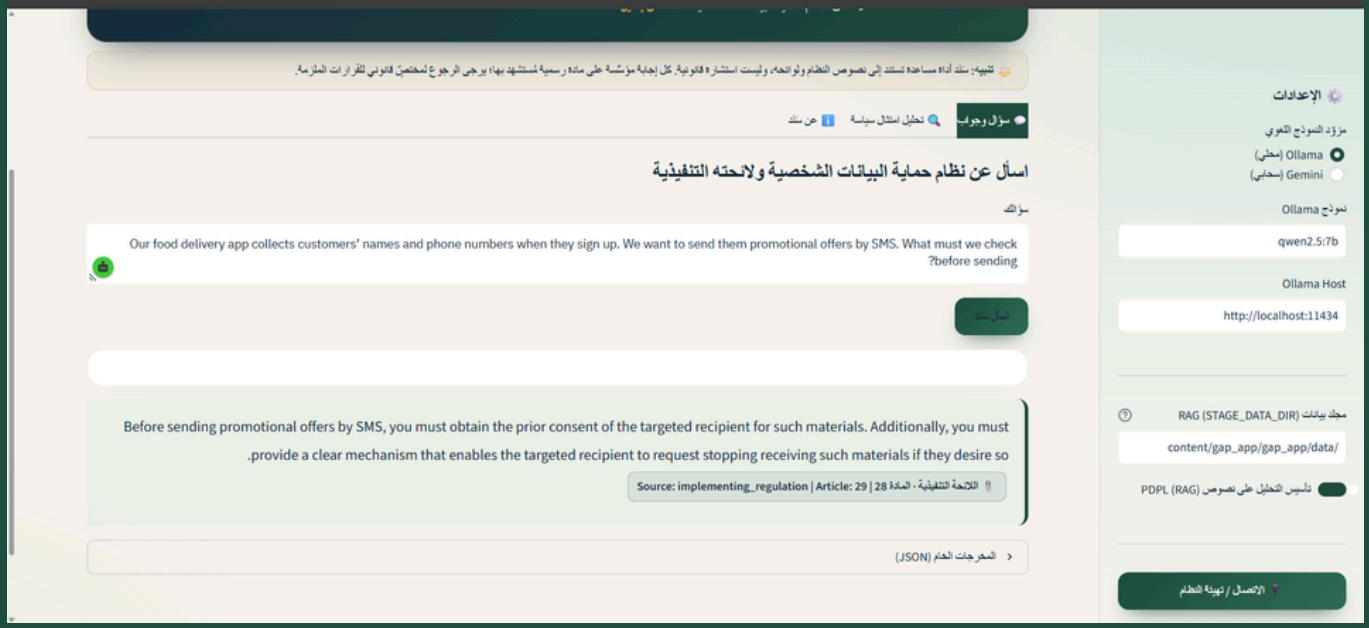
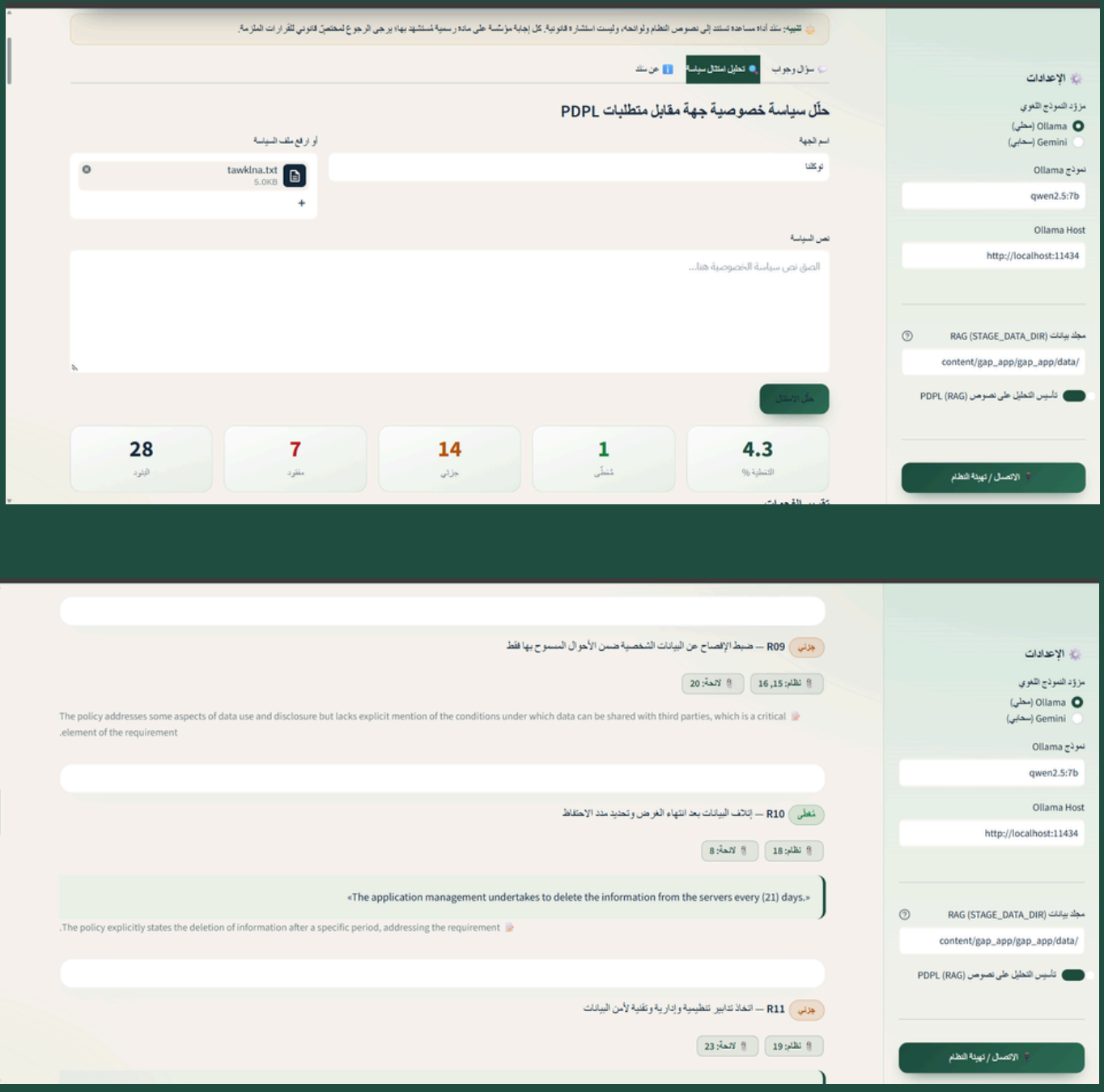


Figure C.3: Gap analysis report for a synthetic policy, showing the three verdict categories (covered, partial, missing) and their counts.



SANAD · سَنَد

Built so that every answer can be checked

سَنَد لا يطلب ثقتك. يُريك دليله.